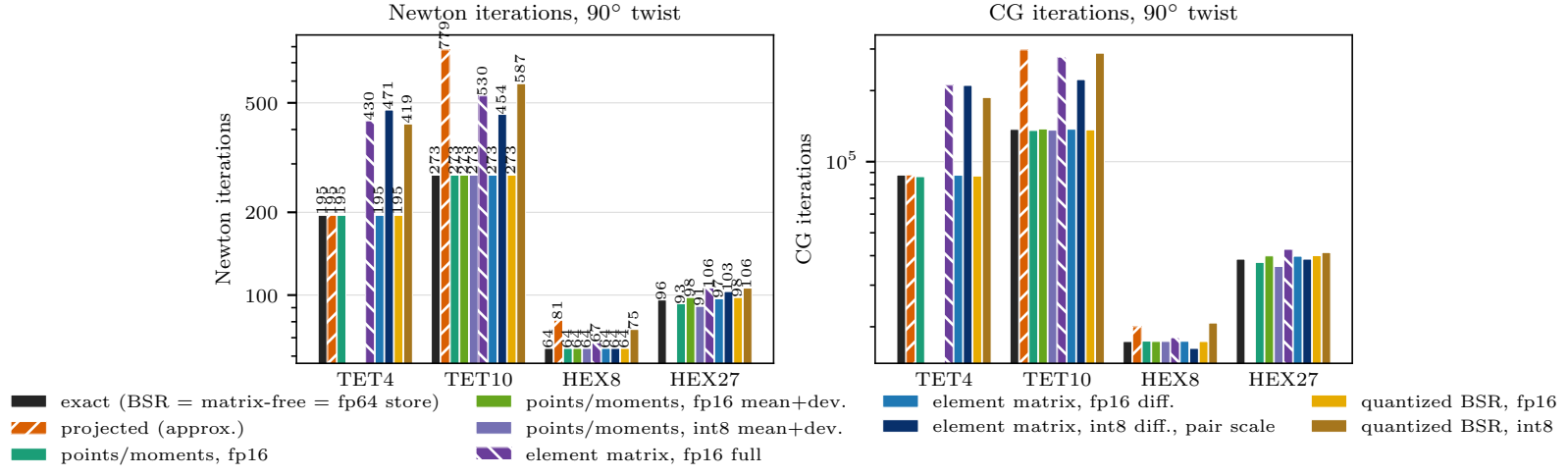

Newton iterations, 90° twist
Newton iterations
500
200
100
195
195
195
430
195
471
195
419
273
779
273
273
273
530
273
454
273
587
64
81
64
64
67
64
64
64
75
96
93
98
97
106
97
103
98
106
TET4
TET10
HEX8
HEX27
CG iterations, 90° twist
CG iterations
10^5
TET4
TET10
HEX8
HEX27
exact (BSR = matrix-free = fp64 store)
projected (approx.)
points/moments, fp16
points/moments, fp16 mean+dev.
points/moments, int8 mean+dev.
element matrix, fp16 full
element matrix, fp16 diff.
element matrix, int8 diff., pair scale
quantized BSR, fp16
quantized BSR, int8